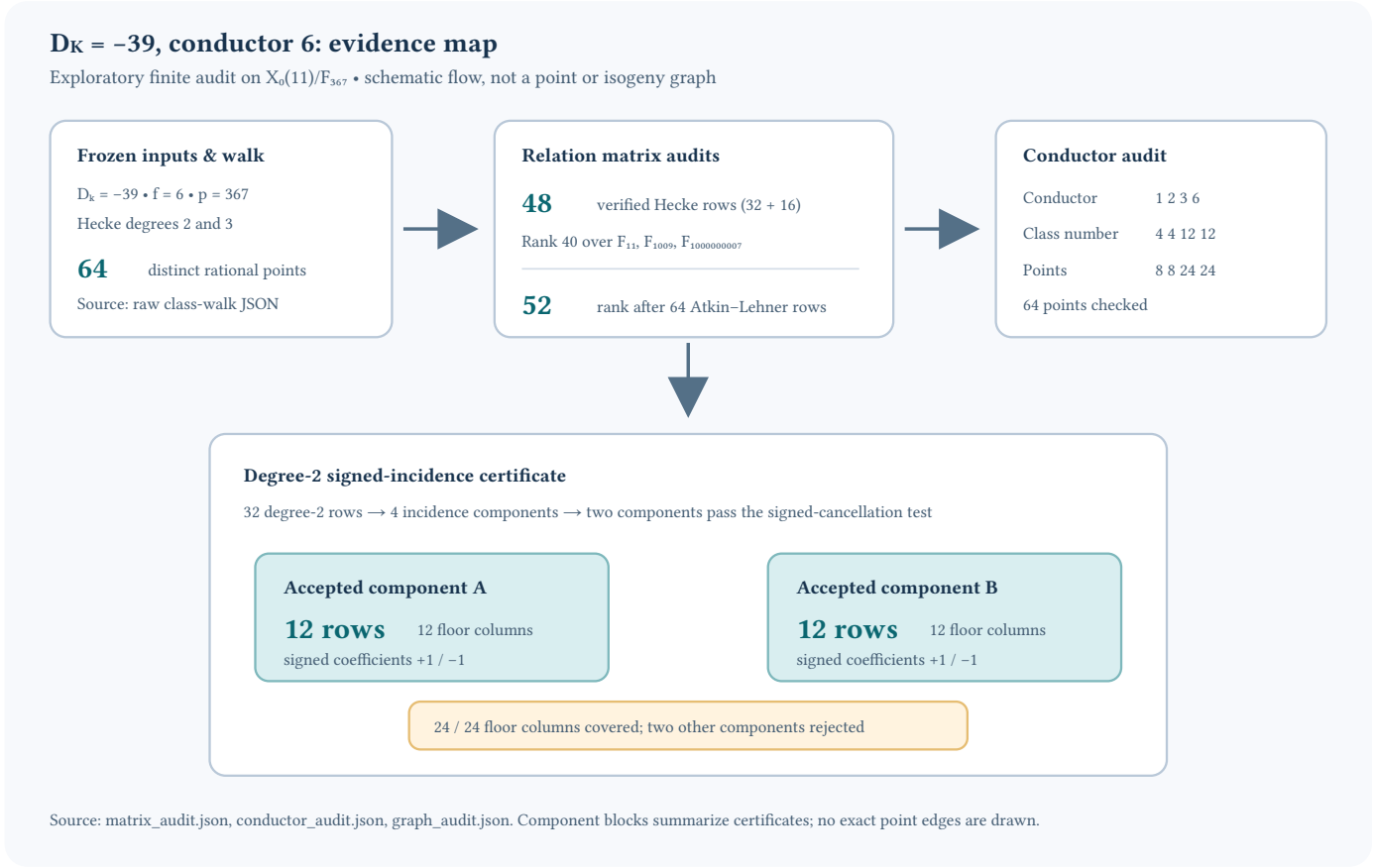
$D_k = -39$ Hecke rank audit

Evidence-linked visual record • exploratory finite case • 2026-10-28

Provenance. This report summarizes already-published run and audit artifacts; it does not repeat the experiment. `PROTOCOL.md` and the result first appear together in the PR, so this is an exploratory comparison, not an independently verifiable holdout. The figure is an audit-flow schematic, not a literal graph of points or isogenies.



Recorded observations

The exploratory class walk on $X_0(11)/F_{367}$, with $D_k = -39$, conductor $f = 6$, and Hecke degrees 2 and 3, records 64 distinct rational points. The matrix audit verifies 32 degree-2 and 16 degree-3 rows. Its sparse rank checks report Hecke rank 40, rising to 52 after 64 Atkin–Lehner rows, consistently over F_{11}, F_{1009}, and $F_{1000000007}$.

The independent conductor audit assigns the 64 points across conductors 1, 2, 3, and 6 as 8, 8, 24, and 24 points. The graph audit starts from 32 degree-2 rows and records four incidence components. Two satisfy the signed-cancellation certificate, each with 12 rows; together they cover all 24 conductor-6 floor columns with coefficients +1 or −1. Two components were rejected because row and lower-column counts differ. This certificate does not identify the components with ideal-class orbits; the reported class-action explanation remains conjectural.

Graph-update disposition

We checked the study directory and `docs/ecdlp-surfaces/` visual sources. This case adds a finite signed-incidence audit but does not certify new point-level graph edges or change the established transfer diagram. No canonical graph source or rendered figure is updated. The SVG below is a local evidence-flow schematic only; exact point edges are intentionally omitted.

Evidence links

- RESULT.md — scope, exploratory provenance, recorded results, and limitations.
- PROTOCOL.md — frozen parameters and prediction, published jointly with the result.
- matrix_audit.json — row counts and rank checks; binds raw input SHA-256 `4bcc8eb1905a9ed79a47cc31820e722dc7a6ff87f18bb37aeebafa35e6cc6df9`.
- conductor_audit.json — verified conductor census; binds the same raw input hash.
- graph_audit.json — signed incidence certificate; binds raw input and conductor audit hashes.
- class_order_audit.json — checked order 12 for form (2, 1, 44) in discriminant −351; its connection to rational-lift components is still conjectural.
- launch.json and runtime-info.json — original launch receipt and runtime facts.

All numerical values above are transcribed from the linked JSON records; no values were newly measured or recomputed for this report. The source data and prior result records remain unchanged.